

High-power rhombohedral- $\text{Fe}_2(\text{SO}_4)_3$ with outstanding cycle-performance as Fe-based cathode for K-ion batteries

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ABSTRACT

We report rhombohedral- $\text{Fe}_2(\text{SO}_4)_3$ as a promising cathode material for potassium-ion batteries. Detailed structure information for rhombohedral- $\text{Fe}_2(\text{SO}_4)_3$ was obtained using X-ray diffraction with Rietveld refinement. Based on the structural information, possible atomic sites for K ions and ion diffusion pathway in the structure were determined using bond-valence energy landscape analyses. At C/20 ($1\text{C} = 112\text{ mA g}^{-1}$), rhombohedral- $\text{K}_x\text{Fe}_2(\text{SO}_4)_3$ delivered a specific capacity of $\sim 100\text{ mAh g}^{-1}$ an average operation voltage of $\sim 3.3\text{ V}$ (vs. K^+/K), corresponding to reversible de/intercalation of $\sim 1.78\text{ mol K}^+$ ions. Moreover, even at 5C, rhombohedral- $\text{K}_x\text{Fe}_2(\text{SO}_4)_3$ exhibited capacity retention of $\sim 80\%$ of the capacity measured at C/20, indicating the outstanding power-capability. After 300 cycles at 2C, rhombohedral- $\text{K}_x\text{Fe}_2(\text{SO}_4)_3$ maintained $\sim 83\%$ of its initial specific capacity with high Coulombic efficiency of more than 99%. *Operando* XRD analyses revealed that the XRD peaks of rhombohedral $\text{K}_x\text{Fe}_2(\text{SO}_4)_3$ monotonously shifted during K^+ de/intercalation, indicating a single phase reaction. First-principles calculation confirmed the good agreement between the theoretical properties of rhombohedral- $\text{K}_x\text{Fe}_2(\text{SO}_4)_3$ and the experimental results, including the average operation voltage, power-capability, and phase reaction.

1. Introduction

With increasing environmental concerns over air-pollution and the accumulation of fine dust particles, clean and sustainable energy sources have attracted great attention, and the development of efficient energy storage systems (ESSs) is considered one of the most important issues in the world [1–3]. Currently, Li ion batteries (LIBs) are widely used as ESSs for not only small electronic devices but also grid-scale applications because of their high energy densities [4–6]. However, the explosive growth of demands for ESSs as well as the limited Li resources in the earth may result in increasing production costs of LIBs, which is considered a major drawback of LIBs, preventing their further grid-scale application [7–9]. Many researchers have thus focused on developing new ESSs with low production costs that rely on earth-abundant elements [10].

K-ion batteries (KIBs) have recently received attention as one of the most promising alternative to LIBs owing to their alkali-ion-based reaction mechanism, which is similar to that of LIBs, and the use of K, one of the most earth-abundant resources [11–13]. In addition, the redox potential of K^+/K is -2.93 V (vs. the standard hydrogen electrode (SHE)), which is higher than those of Na^+/Na (-2.71 V vs. SHE) and Mg^{2+}/Mg (-2.27 V vs. SHE), suggesting that KIBs may exhibit larger energy densities

than Na-ion and Mg-ion batteries, which are also considered alternatives to LIBs for grid-scale application [14,15]. The development of novel cathode materials is considered one of the most important issues for KIBs. Layered-type cathode materials in particular have received great attention as promising cathode for KIBs owing to their large gravimetric capacities [16]. However, most layered-type cathode materials for KIBs have low average operation voltages of less than 3.0 V (vs. K^+/K), unlike those for LIBs [17]. Moreover, it has been reported that a deep discharge process to 1.5 V (vs. K^+/K) is required to achieve large specific capacities for layered-type cathode materials for KIBs, which may result in low energy densities of the full-cell system [18]. To develop novel cathode materials with high operation voltage and reasonable specific capacity for KIBs, we have focused on polyanion-type materials [19–21]. The inductive effect derived from the high electronegativity of polyanions has been reported to result in an increase of the redox potential of electrode materials [22,23]. Furthermore, the strong covalent bonds in polyanions result in high structural stability, which prevents the severe structural degradation of electrode materials by repeated de/intercalation of the large K^+ ions [24].

Herein, it is demonstrated that the rhombohedral- $\text{Fe}_2(\text{SO}_4)_3$ ($\text{R-Fe}_2(\text{SO}_4)_3$) exhibits the outstanding electrochemical performances as one of the promising cathode materials for KIBs. The electrochemical

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properties and reaction mechanism of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ in a KIB system are confirmed through combined studies using first-principles calculation and various experiments. At C/20 ($1\text{C} = 113\text{ mA g}^{-1}$), $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ exhibited a specific capacity of $\sim 100\text{ mAh g}^{-1}$ with an average working voltage of $\sim 3.3\text{ V}$ (vs. K^+/K), which corresponds to $\sim 1.78\text{ mol K}^+$ de/intercalation in the $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ structure. Moreover, the specific capacity of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ measured at 5C was reached up to $\sim 80\%$ of the capacity at C/20. After 300 cycles at 2C, the capacity retention of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ was $\sim 83\%$ of the initial capacity, with a Coulombic efficiency of more than 99%. These results indicate the outstanding power-capability and cycle performance of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ as a promising cathode material for KIBs. Furthermore, *Operando* XRD analyses confirmed that the outstanding cycle performance of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ originated from small volume change ($\sim 1.54\%$) during full charge/discharge.

2. Experimental

2.1. Synthesis of $R\text{-Fe}_2(\text{SO}_4)_3$ and carbon-mixing process

$\text{Fe}_2(\text{SO}_4)_3 \cdot n\text{H}_2\text{O}$ (60.0–80.0%, KANTO CHEMICAL) was sintered at $200\text{ }^\circ\text{C}$ for 6 h to remove H_2O . The white $R\text{-Fe}_2(\text{SO}_4)_3$, Super P carbon black and multiwalled carbon nanotube were homogeneously mixed in a molar ratio of 85:14:1, using high-energy ball milling at 200 rpm for 12 h. The ball-milling jar was filled with Ar to prevent water contamination. This sample includes approximately 15 wt% carbon, as confirmed by thermogravimetric analysis (TGA) (Supporting Fig. S1).

2.2. Materials characterization

The $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ was analyzed using XRD (PANalytical, Empyrean) with $\text{Cu K}\alpha$ radiation ($\lambda = 1.54178\text{ \AA}$) in the 2θ range of 10° to 80° with a step size of 0.13° . Rietveld refinement of the XRD data was performed using FullProf software. TGA was performed under an argon atmosphere at a heating rate of $10\text{ }^\circ\text{C min}^{-1}$ using a thermo gravimetric analyzer (Discovery TGA, USA) from $100\text{ }^\circ\text{C}$ to $700\text{ }^\circ\text{C}$. The valence state of Fe in the structure of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ was determined from XANES analysis performed on beamline 10C at the Pohang Accelerator Laboratory (PAL), South Korea. Fe K-edge spectra were collected in transmission mode with an electron energy of 2.5 GeV and a current of 200 mA, and Fe reference spectra were simultaneously obtained from Fe metal foil. Particle size and distribution measurements were performed using SEM (Hitachi S-4700) and high-resolution TEM. The *operando* XRD patterns of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ were obtained using an X-ray diffractometer (PANalytical, Empyrean) with $\text{Cu K}\alpha$ radiation ($\lambda = 1.54178\text{ \AA}$) in the 2θ range of 19° to 35° with a step size of 0.026° . An *Operando* XRD battery cell (ISBC) made by PANalytical with PDC TECH was used for *Operando* XRD experiments. To protect the cell system from water, beryllium (Be) window was applied.

2.3. Electrochemical characterization

The $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ electrode was fabricated with the following composition: 87.5 wt% carbon-mixed $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$, 2.5 wt% Super P carbon black, and 10 wt% polyvinylidene (PVDF) in *N*-methyl-2-pyrrolidone (NMP). The total weight percent of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ in the electrode was 70 wt%. The slurry was applied on Al foil using a doctor blade. The electrode was dried at $100\text{ }^\circ\text{C}$ for 12 h under vacuum oven. Also, the mass loading of the electrode was $\sim 2\text{ mg cm}^{-2}$. CR2032 cells were assembled using a $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ electrode, K metal as the counter electrode, a separator (Whatman GF/F glass fiber), and an electrolyte (0.5 M KPF_6 in an ethylene carbonate (EC) and diethyl carbonate (DEC) in a volume ratio of 50:50). The electrolyte content used for one CR2032 coin-cell is 0.2 ml. The application of the 0.5 M KPF_6 based electrolyte was reported at various research papers on electrode materials for KIBs [25–27]. The coin cells were assembled in an Ar-filled glove box. Galvanostatic charge/discharge tests were performed at various C-rates (C/20,

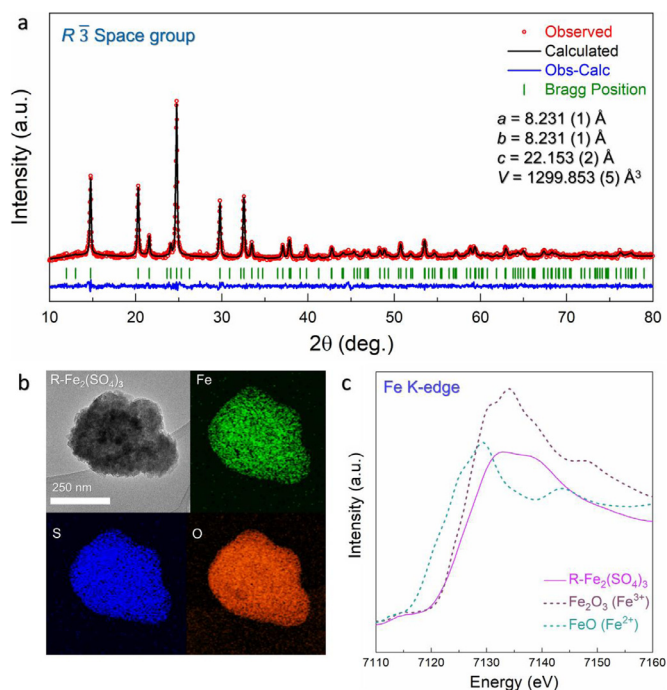


Fig. 1. (a) Refined XRD pattern of $R\text{-Fe}_2(\text{SO}_4)_3$. (b) TEM image of $R\text{-Fe}_2(\text{SO}_4)_3$ and corresponding EDS-mappings on Fe, S and O elements. (c) XANES spectroscopy on Fe K-edge of $R\text{-Fe}_2(\text{SO}_4)_3$ with those of $\text{Fe}^{(3+)}_2\text{O}_3$ and $\text{Fe}^{(2+)}\text{O}$.

C/10, C/5, C/2, 2C and 5C in the range of 2.2–4.1 V, where 2C corresponds to $\sim 79\text{ mA hg}^{-1}$) using an automatic battery charge/discharge test system (WBCS 3000, WonATech).

2.4. Computational details

All the density functional theory (DFT) calculations were performed using the Vienna AB initio Simulation Package (VASP) [28]. The projector-augmented wave pseudopotentials was used with a plane-wave basis set as implemented in VASP [29]. Perdew-Burke-Ernzerhof parametrization of the generalized gradient approximation (GGA) [30] was used for the exchange-correlation functional. The GGA+U method [31] was adopted to address the localization of the d-orbital in Fe ions, using a U value of 4.2 eV from previous studies [32]. An appropriate number of k-points and a kinetic energy cutoff of 500 eV are used in all calculations. All the structures are optimized until the force in the unit cell converges within $0.05\text{ eV \AA}^{-1}$. For the DFT calculations, a $4 \times 4 \times 1$ k-point grid was used to calculate a $1 \times 1 \times 1$ supercell structure of $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$. Using the cluster-assisted statistical mechanics (CASM) software [33], various K/vacancy configurations of each $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ composition were prepared. Of these numerous K/vacancy configurations for each composition, the 20 lowest electrostatic energy configurations for each composition were confirmed using the Ewald summation calculations [34]. NEB calculations [35] were performed to determine the activation barrier for Na diffusion in the $R\text{-K}_x\text{Fe}_2(\text{SO}_4)_3$ structure. To perform the calculations, five intermediate images are generated between each K site. These structures were then calculated using NEB algorithm with the fixed lattice parameters and free internal atomic positions.

3. Results and discussion

To verify the successful preparation of $R\text{-Fe}_2(\text{SO}_4)_3$, XRD analyses with Rietveld refinement were performed. As shown in Fig. 1a, $R\text{-Fe}_2(\text{SO}_4)_3$ was confirmed to exhibit a rhombohedral structure with space group $R\bar{3}c$ without any impurities or second phases. The lat-

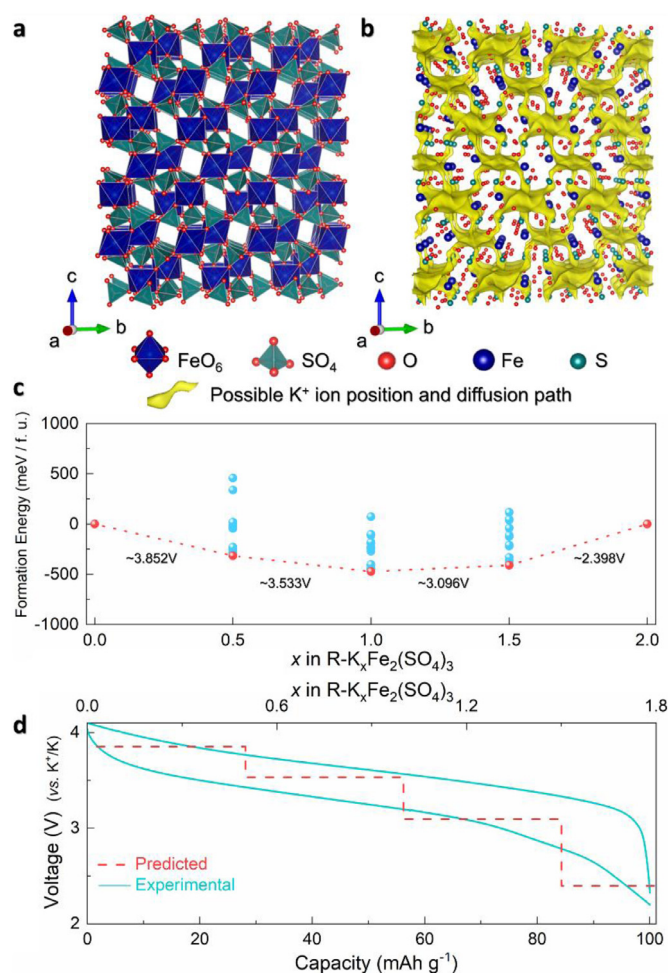


Fig. 2. (a) Crystal structure of R-Fe₂(SO₄)₃ and (b) possible vacant sites and three-dimensional diffusion pathways for K⁺ ions in R-Fe₂(SO₄)₃ structure predicted through BVEL analyses. (c) Convex-hull plot for various R-K_xFe₂(SO₄)₃ configurations (0 ≤ x ≤ 2) with theoretical redox potential. (d) Comparison of predicted redox potential of R-K_xFe₂(SO₄)₃ and its experimentally measured charge/discharge curves for the cycle.

tice parameters of R-Fe₂(SO₄)₃ were verified to be $a = 8.23130(16)$ Å, $b = 8.23130(16)$ Å and $c = 22.1524(7)$ Å; the detailed structural information, such as atomic position, B_{iso} and occupancies, is presented in Supporting Table S1. The low reliable factor ($R_p = 1.16\%$, $R_1 = 1.47\%$, $R_F = 3.87\%$, and $\chi^2 = 0.809\%$) indicate the high accuracy of the Rietveld refinement on R-Fe₂(SO₄)₃. The morphology of R-Fe₂(SO₄)₃ was also investigated using transmission electron microscopy (TEM) analyses combined with energy-dispersive X-ray spectroscopy (EDX) elemental mapping. As shown in Fig. 1b, Fe, S, and O elements were homogeneously distributed in the R-Fe₂(SO₄)₃ particle with an average size of ~400 nm, and the morphology of R-Fe₂(SO₄)₃ was confirmed by scanning electron microscopy (SEM) analysis (Supporting Fig. S2). The elemental ratio of Fe:S in the R-Fe₂(SO₄)₃ particle was ~2: 3, which agrees well with the inductively coupled plasma atomic emission spectroscopy (ICP-AES) analysis (Supporting Table S2). In addition, to verify the oxidation state of Fe ions in R-Fe₂(SO₄)₃, the XANES analyses were performed. As shown in Fig. 1c, the energy level of Fe K-edge on R-Fe₂(SO₄)₃ is similar with that on Fe⁽³⁺⁾₂O₃ rather than that on Fe⁽²⁺⁾O, which indicates that the oxidation state of Fe ions in R-Fe₂(SO₄)₃ is +3.

To predict the vacant site into which K⁺ ions can be intercalated and the possible K⁺ ion diffusion paths, we performed bond-valence energy landscape (BVEL) analyses using the *Bond_str* program in the Fullprof package [36,37]. Fig. 2a and b shows the three-dimensional pathways

for K⁺ diffusion in the R-K₂Fe₂(SO₄)₃ structure, which implies the facile K⁺ diffusion in the structure and the outstanding power-capability of R-Fe₂(SO₄)₃ in the K-cell system. Moreover, it was predicted the Wyckoff position and atomic coordinates (x, y, z) of possible vacant sites for K⁺ ions in the structure, such as (0, 0, 0.5), (0.337, 0.318, 0.318), etc., which indicates that there are numerous vacant sites for K⁺ ions in R-Fe₂(SO₄)₃ structure.

To predict the theoretical properties and possible reaction mechanism of R-K_xFe₂(SO₄)₃ during K⁺ de/intercalation, we performed First-principles calculation based on the structural information obtained from the XRD and the BVEL analyses. The theoretical redox potentials of R-K_xFe₂(SO₄)₃ under the K-cell system were calculated using the following equation:

$$V = - \frac{E[K_xFe_2(SO_4)_3] - E[K_{x-1}Fe_2(SO_4)_3] - (x_2 - x_1)E(K)}{(x_2 - x_1)F} \quad (1)$$

where V is the average redox potential between each K_xFe₂(SO₄)₃, $E[K_xFe_2(SO_4)_3]$ is the formation energy of the most stable configuration K_xFe₂(SO₄)₃ calculated by First-principles calculation. For accurate prediction of theoretical properties of R-K_xFe₂(SO₄)₃, various K⁺/vacancy configurations for R-K_xFe₂(SO₄)₃ (0 ≤ x ≤ 2) were prepared using the cluster-assisted statistical mechanics (CASM) software [33]. Fig. 2c presents a convex-hull plot of the formation energies of R-K_xFe₂(SO₄)₃ as function of K contents in the structure. The results indicate that ~1.78 mol K⁺ ion can be reversibly de/intercalated at R-K_xFe₂(SO₄)₃ in the available voltage range between 2.2 and 4.1 V (vs. K^{+/K}). Moreover, the existence of several stable intermittent phases between R-K₂Fe₂(SO₄)₃ and R-K₀Fe₂(SO₄)₃ was predicted, which implies a single-phase reaction of R-K_xFe₂(SO₄)₃ during K⁺ de/intercalation. In Fig. 2d, the theoretical redox potentials of R-K_xFe₂(SO₄)₃ are compared with the charge/discharge profile of R-K₂Fe₂(SO₄)₃ measured at C/20 (1 C = 112 mA hg⁻¹). It was confirmed that ~1.78 mol K⁺ ions are de/intercalated in the R-K_xFe₂(SO₄)₃ structure with an average operation voltage of ~3.3 V (vs. K^{+/K}). In addition, we tested the electrochemical performances of Fe₂(SO₄)₃·nH₂O to verify effects of crystal water in R-K_xFe₂(SO₄)₃ structure. The XRD pattern of R-Fe₂(SO₄)₃·nH₂O powder, which is clearly different from that of R-Fe₂(SO₄)₃ powder after heat-treatment (Supporting Fig. S3). The R-Fe₂(SO₄)₃·nH₂O electrode was prepared through the same conditions used for the crystal-water-removed R-Fe₂(SO₄)₃ electrode. As presented in Supporting Fig. S4, it was verified that R-Fe₂(SO₄)₃·nH₂O exhibit poor electrochemical performances compared to Fe₂(SO₄)₃ after heat-treatment, which implies that existence of crystal water in the structure results in the negative effect for electrochemical performances of R-Fe₂(SO₄)₃, such as decrease of the theoretical gravimetric capacity resulting from increased molar mass of the active material. In particular, the average operation voltage of R-K_xFe₂(SO₄)₃ is higher than that of other Fe-based cathode materials for KIBs, which is related to previous research on the increased redox potential arising from the ‘inductive effect’ by polyanions such as (PO₄)³⁻ and (SO₄)²⁻. Moreover, R-K_xFe₂(SO₄)₃ exhibits sloppy charge/discharge curves corresponding to a single-phase reaction. Thus, the predictions based on First-principles calculation agrees well with the experimentally verified data.

Electrochemical tests on R-K_xFe₂(SO₄)₃ were performed using CR-2032-type coin cells using the K metal as the counter electrode. Supporting Fig. S5 present the initial charge/discharge capacities of R-K_xFe₂(SO₄)₃ measured at the current rate of C/20 (1 C = 112 mA hg⁻¹) in the voltage range between 2.2 and 4.1 V (vs. K^{+/K}). It was verified that R-K_xFe₂(SO₄)₃ delivered a specific capacity measured at ~100 mAh g⁻¹ at C/20, which indicates ~1.78 mol K⁺ intercalation in R-K_xFe₂(SO₄)₃ structure by electrochemical charge/discharge process. Moreover, up to ~80% of the capacity measured at C/20 was retained even at 5C (Fig. 3a–b), indicating the outstanding power-capability of R-K_xFe₂(SO₄)₃ as a cathode for KIBs. This experimental result was also confirmed by First-principles calculation based on the nudged elastic band (NEB) method. As shown in Fig. 3c–d, the K⁺ diffusion motion

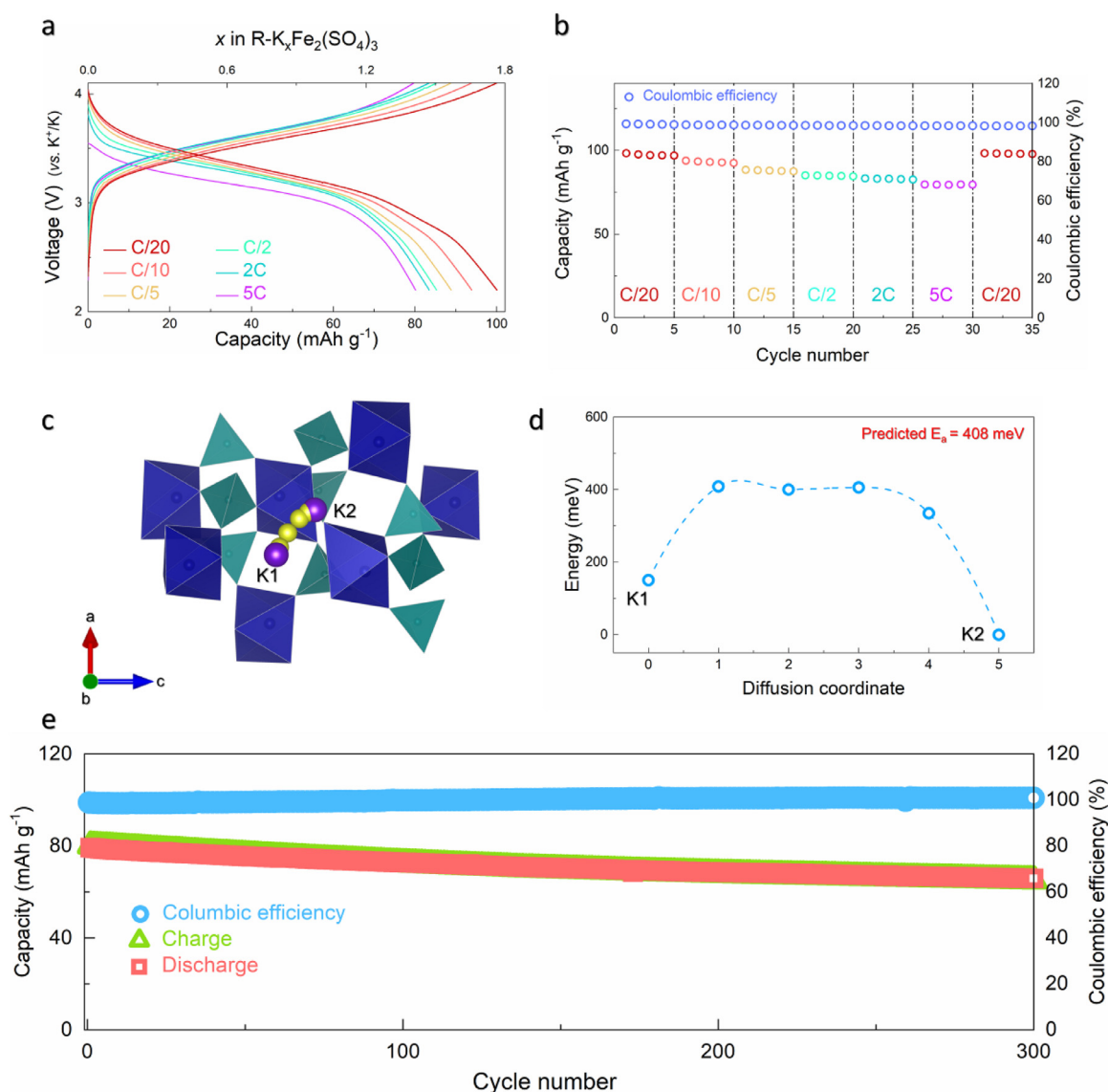


Fig. 3. (a) Charge/discharge profiles of $R-K_xFe_2(SO_4)_3$ in the range of 2.2–4.1 V at various current rates. (b) Power-capability of $R-K_xFe_2(SO_4)_3$ with several cycles at various current rates. (c) Predicted K^+ diffusion motion in $R-K_xFe_2(SO_4)_3$ and (d) predicted activation barrier energy for K^+ diffusion in $R-K_xFe_2(SO_4)_3$. (e) Cycle performance of $R-K_xFe_2(SO_4)_3$ at 2C over 300 cycles in the voltage range of 2.2–4.1 V (vs. K^+/K).

in $R-K_xFe_2(SO_4)_3$ structure during K^+ de/intercalation was predicted in detail, and the activation barrier energy for K^+ diffusion along the $K_{1-1}-K_{2-1}$ pathway was determined to be $\sim 408\ meV$. This low activation barrier energy enables the facile K^+ diffusion in the structure, resulting in the outstanding power-capability of $R-K_xFe_2(SO_4)_3$. In addition, we tested two kinds of the electrolyte with different molar ratio of KPF_6 contents such as 0.5 M and 1 M. As presented in Supporting Fig. S6, it was verified that the electrochemical performances based on 0.5 M KPF_6 -electrolyte are better than those based on 1 M KPF_6 -electrolyte. In particular, in case of the electrochemical performances based on 1 M KPF_6 -electrolyte, the charge capacities are larger than the discharge capacities, indicating the poor Coulombic efficiency under the KIB system using 1 M KPF_6 -electrolyte by the side reaction at the high voltage. This result is well matched with the previously reported research data showing the comparison of electrochemical performances in the electrolyte with various molar ratio of KPF_6 [38]. In addition, the electrochemical performances of $R-K_xFe_2(SO_4)_3$ were compared with those of other cathode materials for KIBs (Supporting Fig. S7), which indicates that $R-K_xFe_2(SO_4)_3$ delivers the outstanding energy density and

power capability as one of the promising cathode materials for KIBs [39–46]. Furthermore, $R-K_xFe_2(SO_4)_3$ delivered capacity retention of $\sim 83\%$ after 300 cycles at 2C with Coulombic efficiency of more than 99% (Fig. 3e). Moreover, we performed the full-cell tests using $R-Fe_2(SO_4)_3$ as a cathode and potassiated hard-carbon as an anode in the voltage range between 2.1 and 4.0 V (Supporting Fig. S8). The capacity ratio of negative and positive electrode is approximately ~ 1.2 . To prepare the potassiated hard-carbon anode, the hard-carbon electrode was discharged to 0.001 V (vs. K^+/K) under the half-cell system with K-metal anode. At the current density of $6\ mA\ g^{-1}$, $R-Fe_2(SO_4)_3$ /Potassiated hard-carbon full cell delivered an initial capacity of $\sim 100\ mAh\ g^{-1}$. Moreover, the full-cell exhibits the capacity retention of $\sim 88\%$ compared to the initial capacity for 200 cycles at the current density of $224\ mA\ g^{-1}$, which indicates the outstanding cycle-performance of the $R-Fe_2(SO_4)_3$ /Potassiated hard-carbon full cell under the KIB system. It is speculate that the great cyclability of $R-K_xFe_2(SO_4)_3$ results from the small and simple structural change of $R-K_xFe_2(SO_4)_3$ during K^+ de/intercalation. In addition, to verify the solubility of $Fe_2(SO_4)_3$ in the electrolyte, $Fe_2(SO_4)_3$ powder (3 g) was stirred in the electrolyte

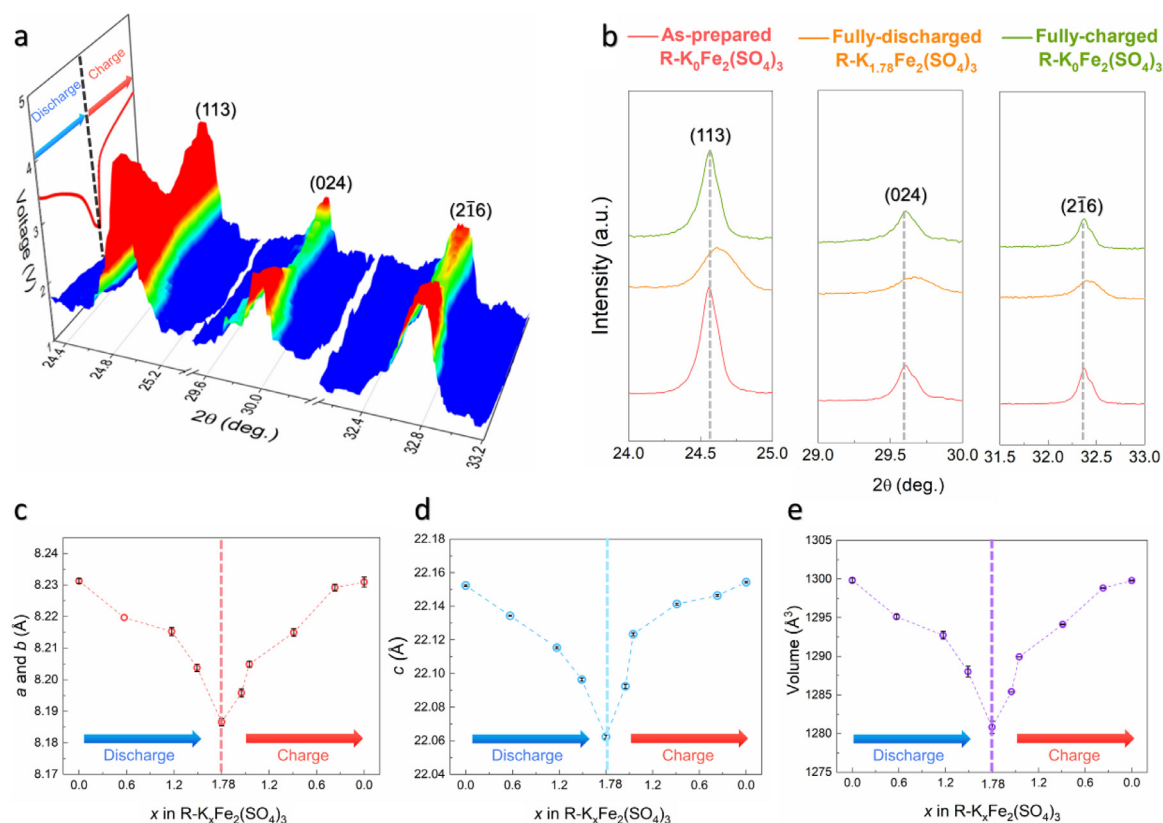


Fig. 4. (a) Operando XRD patterns of $R-K_xFe_2(SO_4)_3$ measured at a current rate of $C/10$ in the voltage range of 2.2–4.1 V (vs. K^+/K). (b) Extended ex-situ XRD patterns of as-prepared $R-K_0Fe_2(SO_4)_3$, fully-discharged $R-K_{1.78}Fe_2(SO_4)_3$ (to 2.2 V) and fully-charged of $R-K_0Fe_2(SO_4)_3$ (to 4.1 V). (c–e) Change in lattice parameters a , b , c and volume of $R-K_xFe_2(SO_4)_3$ during K^+ de/intercalation in the structure.

(10 ml) for 1 week. It was observed that $Fe_2(SO_4)_3$ powder was not dissolved in the EC and DEC based electrolyte. In addition, we performed the Fourier-transform infrared spectroscopy (FTIR) analyses, in order to identify the solubility of $Fe_2(SO_4)_3$ in the electrolyte through detection of S–O bonding. As shown in Supporting Fig. S9, it was verified that the FTIR result of the pristine electrolyte is highly similar with that of the electrolyte in which $Fe_2(SO_4)_3$ powder was stirred for 1 week, and their data are clearly different from the FTIR result of $Fe_2(SO_4)_3$ powder. Moreover, we performed Inductively Coupled Plasma-Atomic Emission Spectrometer (ICP-AES) analysis. As presented in Supporting Table S3 below, it was confirmed that Fe element is not detected in the electrolyte after charge/discharge processes of $R-Fe_2(SO_4)_3$ under the KIB system. Thus, we confirmed that it is difficult to dissolve $Fe_2(SO_4)_3$ powder in the EC: DEC based electrolyte.

Fig. 4a presents Operando XRD patterns of $R-K_xFe_2(SO_4)_3$ during charge/discharge at $C/10$ ($1\text{ C} = 112\text{ mA hg}^{-1}$) in the voltage range between 2.2 and 4.1 V (vs. K^+/K). The changes of the main peaks indicate that $R-K_xFe_2(SO_4)_3$ underwent a single-phase reaction during K^+ de/intercalation. The peak intensities in the Operando XRD patterns are also reversibly changed during charge/discharge, which indicates the reversible structural change of $R-K_xFe_2(SO_4)_3$ by K^+ de/intercalation. In particular, the XRD peaks corresponding to (113), (024) and (216) planes of $R-K_xFe_2(SO_4)_3$ crystal structure underwent small monotonous shifts during charge/discharge, as confirmed by ex-situ XRD analyses of $R-K_xFe_2(SO_4)_3$ (Fig. 4b and Supporting Fig. S10). It was confirmed that the lattice parameters and volume of $R-K_xFe_2(SO_4)_3$ is decreased (or increased) during K^+ intercalation (or deintercalation), corresponding to shift of the XRD peaks toward higher (or smaller) 2θ angles during discharge (or charge) (Fig. 4c–e). It was verified that the total volume changes of $R-K_xFe_2(SO_4)_3$ during charge/discharge was just $\sim 1.54\%$. Thus, it is speculated that the small structural change of $R-K_xFe_2(SO_4)_3$

during K^+ de/intercalation results in the outstanding cycling performance of $R-K_xFe_2(SO_4)_3$ as a promising cathode for KIBs. In addition, through the SEM, TEM and XRD analyses, it was confirmed that the overall morphology and crystal structure of $R-K_xFe_2(SO_4)_3$ were well retained even after 300 cycles (Supporting Fig. S11), which also contributes to the excellent cycle performance of $R-K_xFe_2(SO_4)_3$ in the KIB system. In addition, the Rietveld refinement on the full discharged $R-K_xFe_2(SO_4)_3$ was performed to confirm the atomic position of intercalated K^+ ions in the structure (Supporting Fig. S12 and Supporting Table S4). The Wyckoff position and atomic positions for the intercalated K^+ ions are as follow; 3b (0, 0, 0.5) for K1 site and 18f (0.337, 0.318, 0.3185) for K2 site, which is well matched with the structural information of the site for K^+ ions predicted through BVLE analyses. Moreover, the Rietveld refinement result indicates that the molar ratio on K, Fe, S and O atoms in the full discharged $R-K_xFe_2(SO_4)_3$ is $\sim 1.78: 2: 3: 12$, which is well matched with the ICP-AES analysis results on the full discharged $R-K_xFe_2(SO_4)_3$.

To investigate the reaction mechanism of $R-K_xFe_2(SO_4)_3$ during charge/discharge, such as the change of the oxidation state of Fe ions, we performed the ex-situ XANES analyses. As shown in Fig. 5a, the Fe K-edge was shifted to lower energy level during discharge to 2.2 V (vs. K^+/K), indicating that the Fe^{2+}/Fe^{3+} redox reaction occurs with 1.78 mol K^+ intercalation in $R-K_xFe_2(SO_4)_3$. The Fe^{2+}/Fe^{3+} redox reaction in $R-K_xFe_2(SO_4)_3$ was also confirmed through First-principles calculation. Fig. 5b presents the integrated spin moments of Fe ions on $R-K_0Fe_2(SO_4)_3$ and $R-K_{1.5}Fe_2(SO_4)_3$. During K^+ intercalation between $R-K_0Fe_2(SO_4)_3$, the total electron spin counts of Fe ions changed from +5 to +4 of $R-K_xFe_2(SO_4)_3$, implying that Fe^{3+} ions in $R-K_0Fe_2(SO_4)_3$ are transformed into Fe^{2+} ions with K^+ intercalation in the structure. Fourier transform (FT) analyses on the ex-situ EXAFS data of $R-K_xFe_2(SO_4)_3$ revealed that Fe–O bonding distance in $R-K_xFe_2(SO_4)_3$ structure is contin-

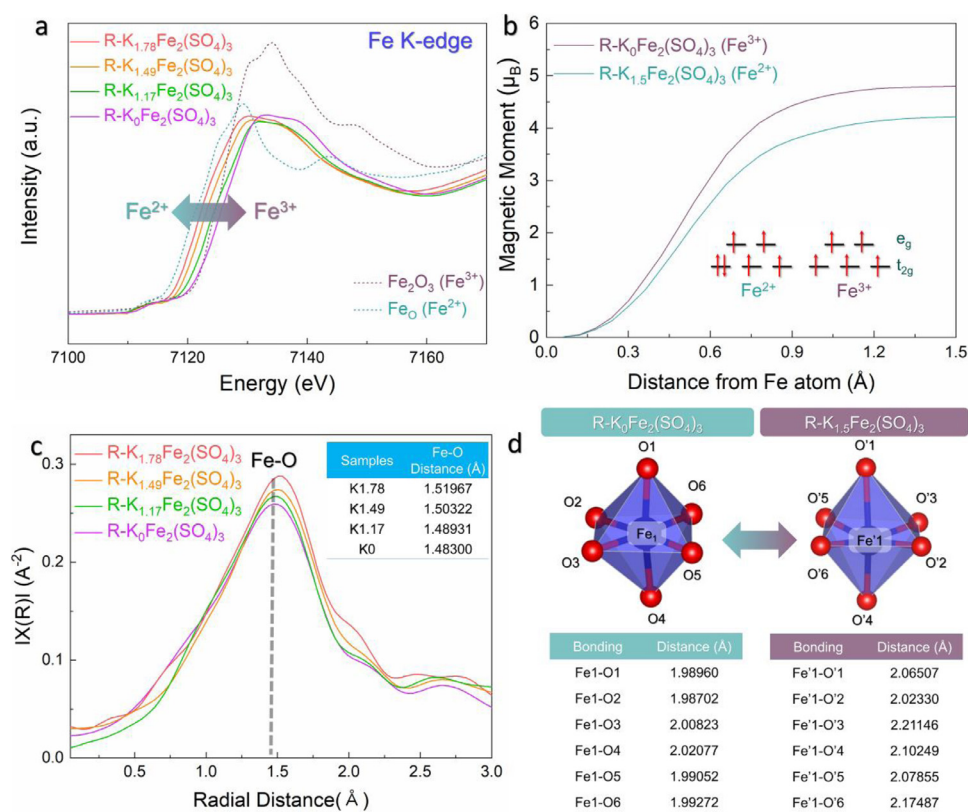


Fig. 5. (a) *Ex-situ* Fe K-edge XANES analyses of $R-K_xFe_2(SO_4)_3$ ($0 \leq x \leq 1.78$). (b) Comparison of integrated spin moments on Fe ion between $R-K_0Fe_2(SO_4)_3$ and $R-K_{1.5}Fe_2(SO_4)_3$. (c) *Ex-situ* EXAFS analyses of $R-K_xFe_2(SO_4)_3$ ($0 \leq x \leq 1.78$) with the Fe-O bonding distance. (d) Comparison of predicted Fe-O bond distances of the FeO_6 octahedra between $R-K_0Fe_2(SO_4)_3$ and $R-K_{1.5}Fe_2(SO_4)_3$ through first-principles calculation.

uously and slightly increased and decreased during K^+ de/intercalation by change of oxidation state between Fe^{3+} and Fe^{2+} (Fig. 5c), which is also confirmed through First-principles calculation results predicting change of the Fe-O bonding environment in the FeO_6 octahedra between $R-K_0Fe_2(SO_4)_3$ and $R-K_{1.5}Fe_2(SO_4)_3$ predicted by First-principles calculation (Fig. 5d). In addition, to increase the specific capacity of cathode materials for KIBs, it is required that the K^+ storage per the molar unit of cathode materials becomes larger, which implies that the more electrons used for K^+ de/intercalation should be provided from the cathode materials. Thus, we guess that increasing the specific capacity of cathode materials for KIBs can be enabled through application of cationic-redox reaction based multi-electron reaction by transition metal or anion-redox reaction by oxygen.

4. Conclusion

We demonstrated that $R-K_xFe_2(SO_4)_3$ exhibits the outstanding electrochemical performances as a promising cathode for KIBs. Through combined studies using First-principles calculation and various experiments, the theoretical properties of $R-K_xFe_2(SO_4)_3$ in the KIB system are confirmed, including the possible atomic sites for K^+ ions, the redox potentials, and the activation barrier energies for K^+ diffusion and the Fe^{2+}/Fe^{3+} redox reaction. At C/20 ($1\text{ C} = 112\text{ mA h g}^{-1}$), $R-K_xFe_2(SO_4)_3$ delivered a specific capacity of $\sim 100\text{ mA h g}^{-1}$, corresponding to the intercalation of 1.78 mol K^+ with an average operation voltage $\sim 3.3\text{ V}$ (vs. K^+/K). Even at 5C, up to $\sim 80\%$ of the specific capacity measured at C/20 was retained, demonstrating the outstanding power-capability of $R-K_xFe_2(SO_4)_3$ under the KIB system. Furthermore, over 300 cycles at 2C, $R-K_xFe_2(SO_4)_3$ exhibited capacity retention of $\sim 83\%$ with high Coulombic efficiency of more than 99%, resulting from the small volume change ($\sim 1.54\%$) during de/intercalation of K^+ ions, as confirmed by *Operando* XRD analyses. We believe that our findings provide great insight into the applicability of various materials as promising electrodes for not only KIBs but also other rechargeable batteries.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jungmin Kang: Conceptualization, Validation, Investigation, Resources, Visualization. **Wonseok Ko:** Visualization, Formal analysis, Methodology. **Hyunyoung Park:** Formal analysis, Validation. **Yongseok Lee:** Investigation. **Jae Hyeon Jo:** Validation. **Ji Ung Choi:** Conceptualization. **Seung-Taek Myung:** Conceptualization. **Jongsoo Kim:** Project administration, Supervision, Visualization, Funding acquisition.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ensm.2020.08.024.

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